

Iteration Plan (Unified Process)

Weeks	UP Phase	Milestone	Status
1–2	Inception	Vision & Feasibility	Complete
3–5	Elaboration (Iteration 1)	Core Architecture & Requirements	Complete
6–8	Elaboration (Iteration 2)	Risk-Driven Design	Complete
9–12	Construction (Iterations 3–4)	Feature Implementation	Complete
13–14	Transition (Iteration 5)	Polishing & Deployment Prep	Complete
15	Transition	Demo & Final Submission	Not started

Iteration 1 — Inception (Weeks 1–2)

Objectives

- Define project scope
- Identify stakeholders
- Develop high-level requirements
- Identify major risks
- Prepare initial project documentation

Deliverables

- Business case
- High-level requirements
- Use case list (8 high-level use cases)
- Risk list
- Iteration plan

Iteration 2 — Elaboration, Part 1 (Weeks 3–5)

Objectives

- Detail ~30% of use cases (key/high-risk scenarios), with system sequence diagrams
- Build the domain model (conceptual class diagram)
- Produce an architectural proof-of-concept (layered design)
- Update the risk list with mitigation strategies

Deliverables

- `detailed_use_cases_iteration1.md` — 9 detailed use cases with SSDs
- `domain_model.md` — conceptual class diagram write-up
- `layered_architecture.md` — reconciled layered architecture (UI / Domain / Business Infrastructure / Technical Services)
- `risk_list.md` — updated with likelihood/impact/mitigation

Iteration 3 — Elaboration, Part 2 (Weeks 6–8)

Objectives

- Detail the remaining ~70% of use cases, with system sequence diagrams
- Refine the domain model into a design class diagram (with methods)
- Build a UI prototype
- Track risk mitigation progress

Deliverables

- `detailed_use_cases_iteration2.md` — remaining 9 detailed use cases with SSDs
- `design_class_diagram.md` — design classes with methods (GRASP patterns applied)
- `ui_prototype.md` + HTML mockup — Login, Sales/Checkout, Inventory, Reporting screens
- `risk_list.md` — mitigation progress/status column added

Iteration 4 — Construction (Weeks 9–12)

Objectives

- Implement prioritized features per the design: sales processing, inventory management, reporting
- Write unit and integration tests
- Set up version control / continuous integration
- Track defects

Planned Deliverables

- Working code for the layered architecture (Python backend behind a REST API, React front end)
- Unit + integration test suite
- Git repository with commit history
- Defect tracking log

Iteration 5 — Transition (Weeks 13–14)

Objectives

- Beta test with representative users/scenarios and collect feedback
- Performance and security tuning
- Finalize documentation (updated models, user manual)

Planned Deliverables

- Beta testing report
- Performance/security fix log
- User manual + final updated models

Final Submission (Week 15) Not started

Planned Deliverables

- Live demo of key scenarios
- Final report (UP process reflection, lessons learned)
- Complete code & documentation repository